NOTRE DAME OF MIDSAYAP COLLEGE

College of Information Technology and Engineering • Bachelor of Science in Computer Science

SYSTEM PROTOTYPE SCREENSHOTS & DESCRIPTIONS

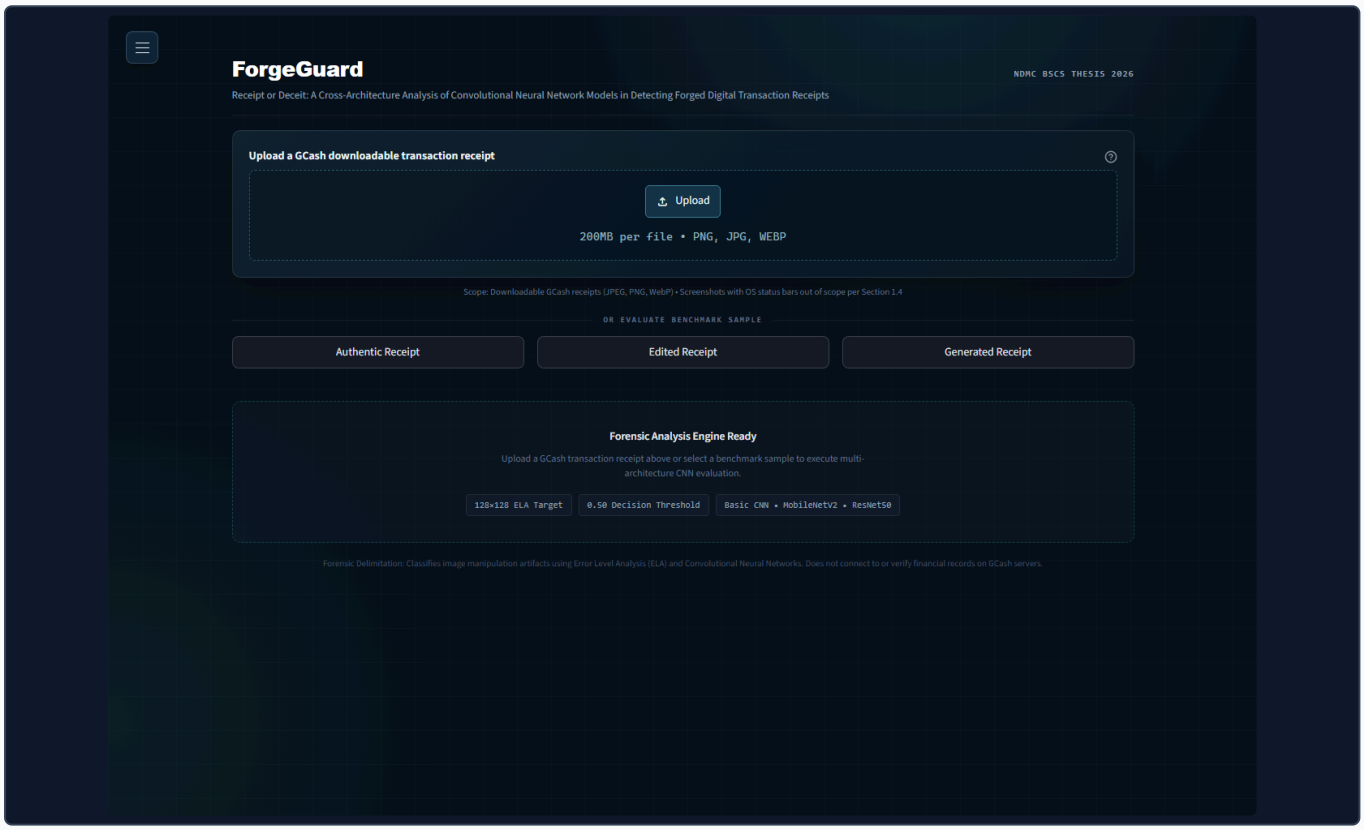
Approved Canonical Title: "Receipt or Deceit: A Cross-Architecture Analysis of Convolutional Neural Network Models in Detecting Forged Digital Transaction Receipts"

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Adviser:	Ms. Doris Ann Mariano	**Folder:**	01_System_Prototype / 06_Screenshots_with_Descriptions

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EXHIBIT 01 System Landing Dashboard & Intake Zone

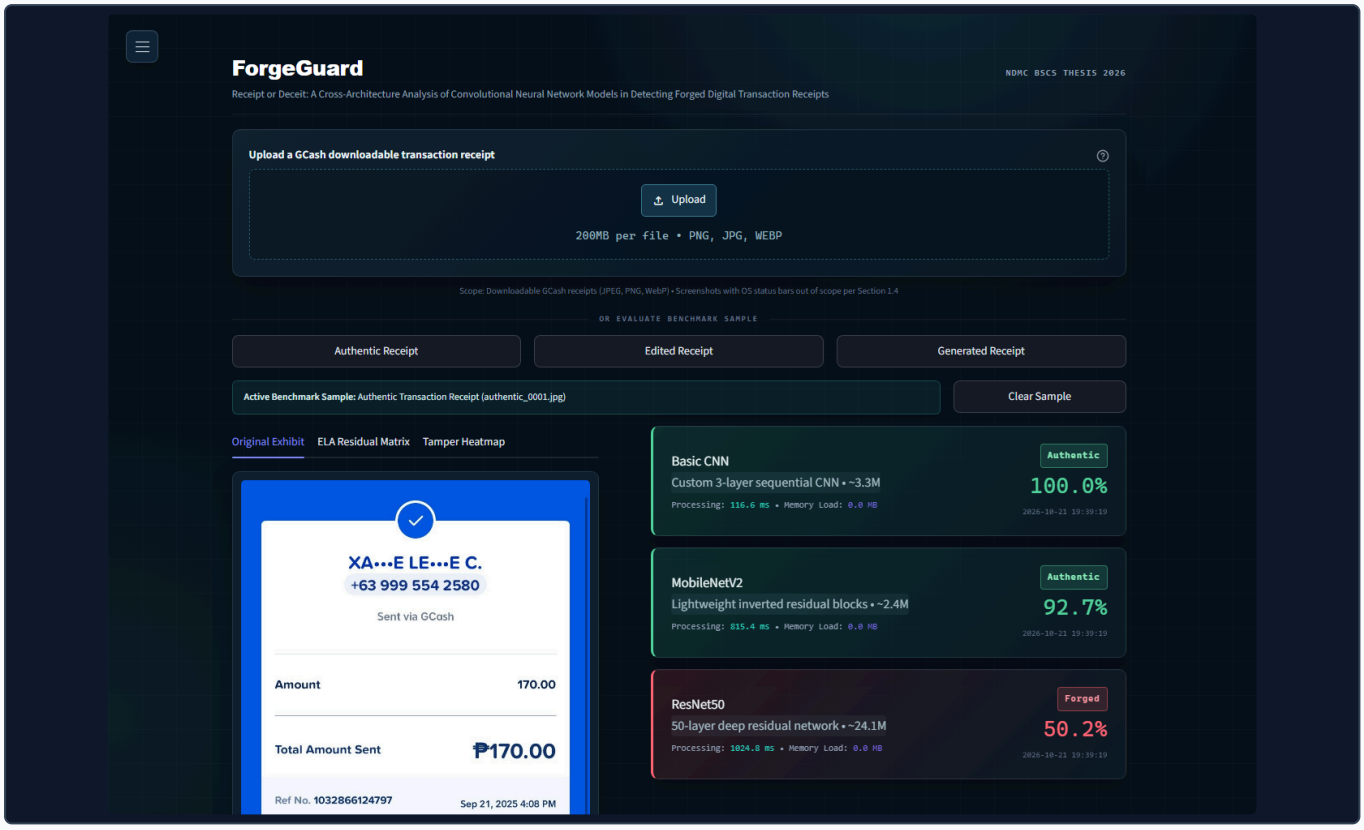
Figure 4, Layer 1 (User Interface Layer)



Technical Description: The primary operational interface of ForgeGuard. Supports intake of high-resolution downloadable GCash transaction slips in JPEG, PNG, and WebP formats up to 200MB. Interactive one-click benchmark buttons (Authentic Receipt, Edited Receipt, Generated Receipt) allow immediate empirical testing without requiring local file uploads. Technical status badges indicate 128x128 ELA input dimensions, 0.50 sigmoid decision threshold, and active multi-CNN graph compilation.

EXHIBIT 02 **Authentic Receipt Multi-CNN Verification**

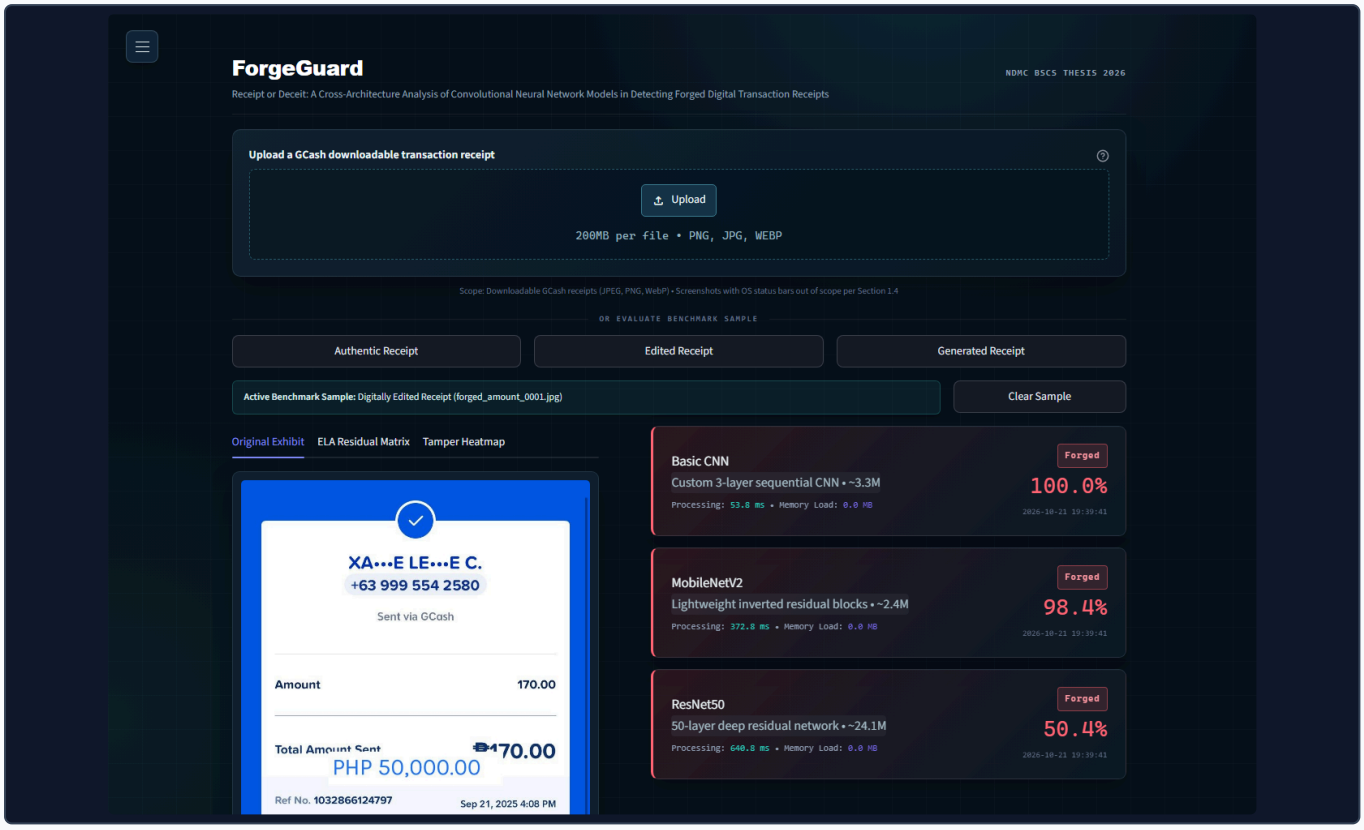
Figure 4, Layers 2 & 3 (Inference Engine & Output Layer)



Technical Description: Diagnostic evaluation of a genuine downloadable GCash payment confirmation slip. Basic CNN verifies authenticity with 100.0% confidence in 116.6 ms. MobileNetV2 independently corroborates authentic status with 92.7% confidence in 815.4 ms. ResNet50 establishes baseline sensitivity at 50.2% fraud verdict, documenting transfer-learning sensitivity to normal JPEG quantization variance.

EXHIBIT 03 **Digitally Edited Receipt Forgery Detection**

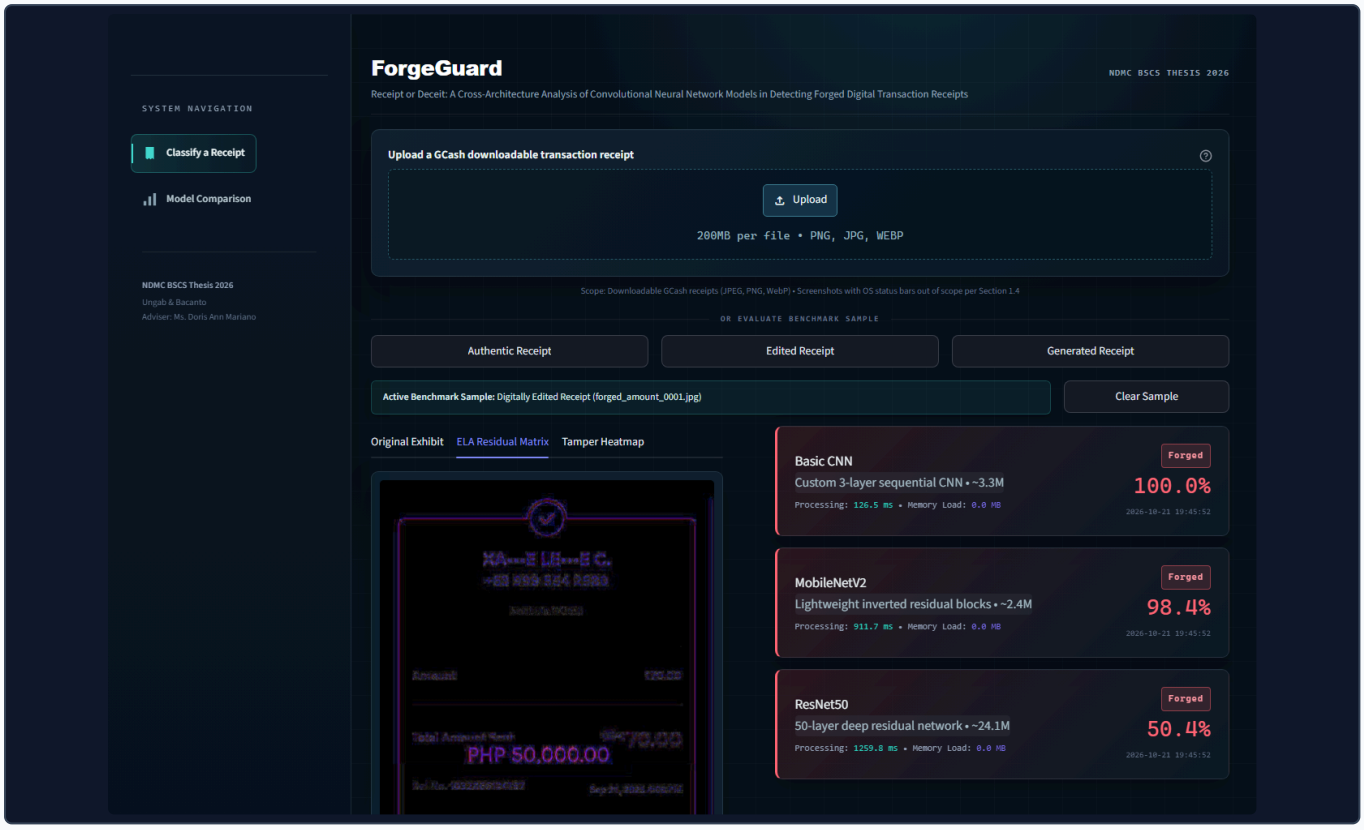
Figure 4, Layer 3 (Result Processing & Fraud Alerting)



Technical Description: Forensic evaluation of an altered transaction slip where amount fields were spliced using image editing software. Basic CNN successfully flags the forgery at 100.0% fraud confidence (latency: 126.5 ms). MobileNetV2 detects the tamper with 98.4% confidence (latency: 911.7 ms). High-contrast security alert cards display instantaneous diagnostic feedback.

EXHIBIT 04 **Error Level Analysis (ELA) Residual Decomposition**

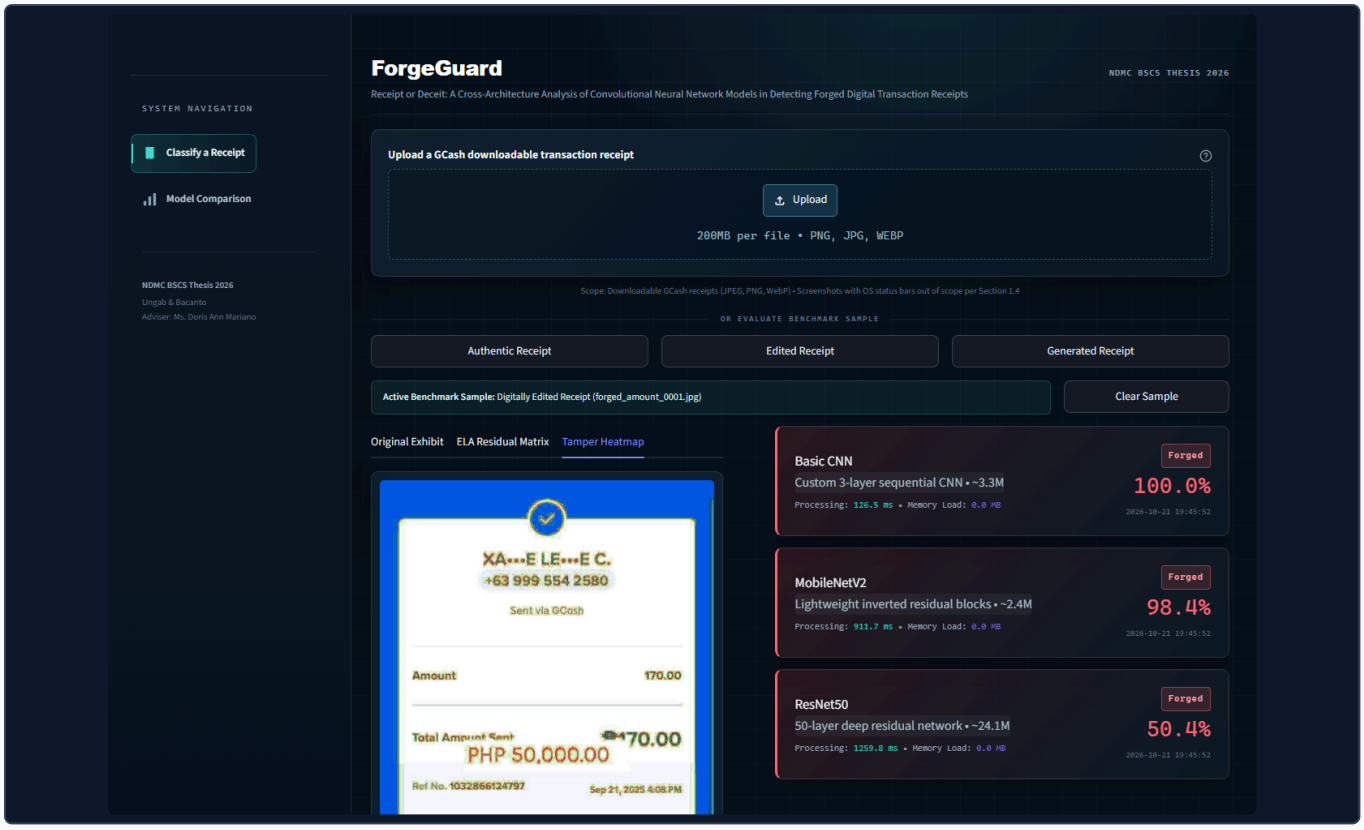
Figure 4, Layer 2 (Image Preprocessing Module - Sections 1.3 & 2.6)



Technical Description: Mathematical decomposition of the receipt canvas generated via calibrated Error Level Analysis at quality Q=90 with 15.0x error amplification. Spliced text regions exhibit distinct compression error spikes due to divergent compression histories against the surrounding background canvas, exposing localized re-compression boundaries.

EXHIBIT 05 **Dynamic Thermal Heatmap Localization**

Figure 4, Layer 4 (Forensic Explainability & Audit Layer)



Technical Description: Color-mapped spatial variance thermal overlay synthesized directly from ELA residual gradients. Unaltered receipt regions remain neutral and unobstructed, while mathematically anomalous compression boundaries surrounding forged amount and reference text glow in vivid orange-red thermal contours, providing human auditors with visual localization.

EXHIBIT 06 **Overall Architecture Benchmark Matrix**

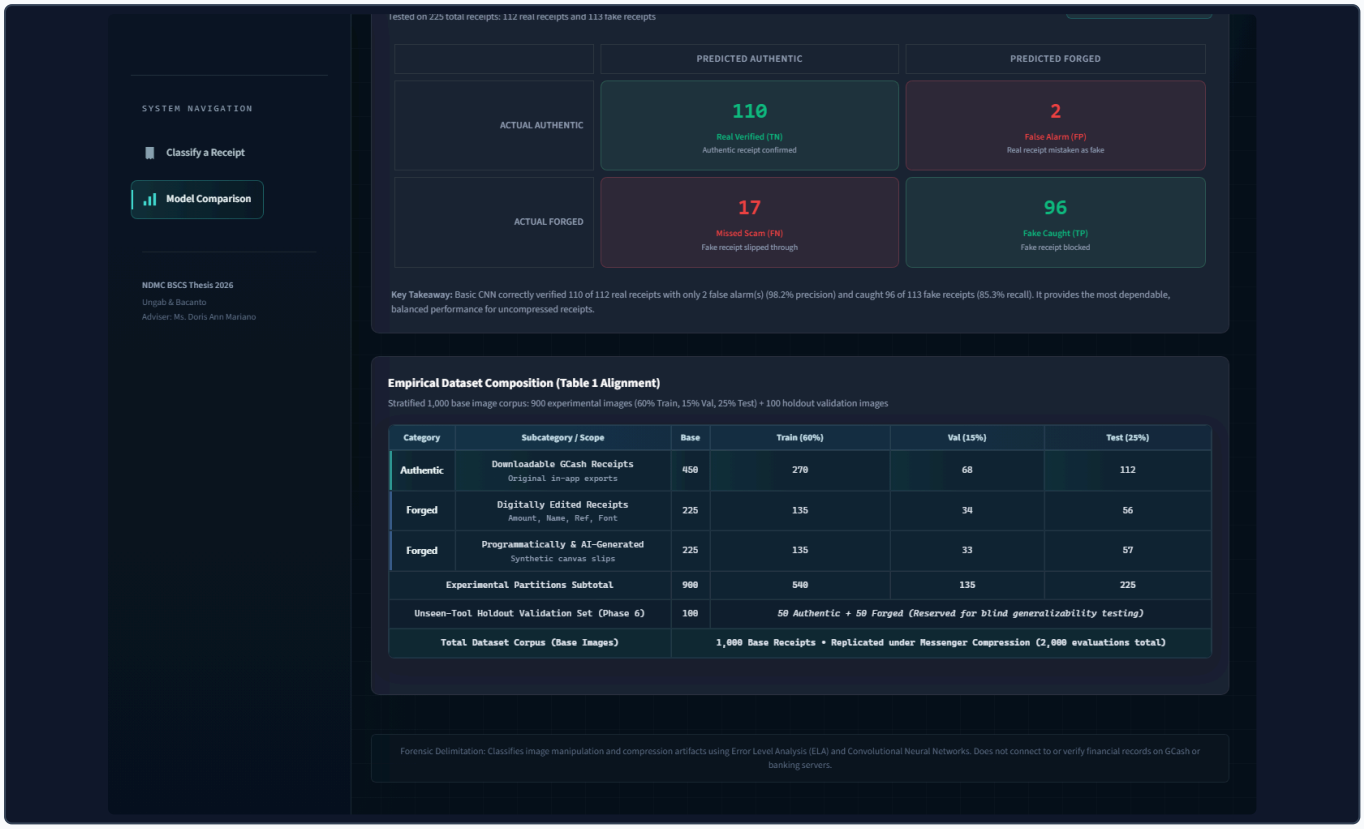
Chapter 2 / Table 2 Comparative Architecture Benchmark



Technical Description: Comprehensive cross-architecture benchmark matrix evaluating Basic CNN, MobileNetV2, and ResNet50 across both Standard and Messenger-compressed conditions. Details Accuracy, Compression Delta (Delta Acc), Precision, Recall, F1-Score, Execution Latency, Peak Working RAM, Parameter Count, and Section 2.10 Multi-Criteria Composite Scores (Basic CNN: 100.00%; MobileNetV2: 71.50% / 66.85%; ResNet50: 0.00%). Centered presentation guarantees balanced data readability.

EXHIBIT 07 **Empirical Dataset Composition Matrix (Table 1 Alignment)**

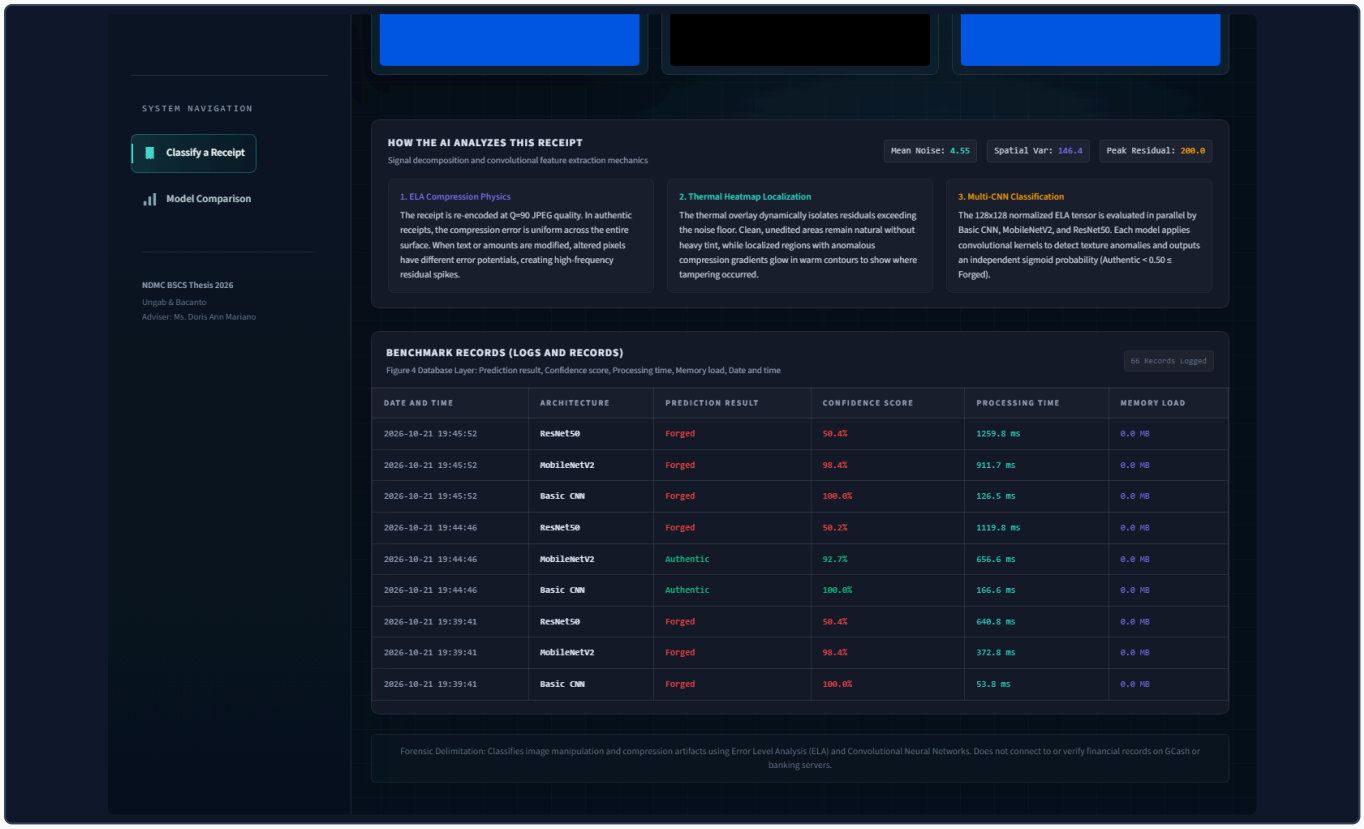
Chapter 1 / Section 1.4 & Table 1 Dataset Stratification



Technical Description: Fully centered layout of the 1,000-image base receipt corpus: 900 experimental images (60% Training: 540 images; 15% Validation: 135 images; 25% Testing: 225 images) and 100 unseen-tool holdout validation images (50 Authentic + 50 Forged). Documents both Standard and Messenger-compressed evaluation conditions (2,000 evaluations total).

EXHIBIT 08 **Benchmark Records Telemetry & Audit Log**

Figure 4, Layer 4 (Database Layer - Section 2.5.3)



Technical Description: Persistent transaction ledger recording Date and Time, Architecture Evaluated, Prediction Result, Confidence Score, Processing Time in milliseconds, and System Memory Load in Megabytes. Proves system compliance with the Database Layer requirement of Figure 4.